Basics

- All states live in a Hilbert space of vectors where $|\psi\rangle$ captures the full system state.
  ⇒ State vectors obey superposition.
- Every dynamical variable has a HERMITIAN operator
- Measurement yields the e-value of the operator.
  Expectation value is $\langle\psi|A|\psi\rangle = a\cdot\langle\psi|\psi\rangle = a$
- Time evolution is given by Schrödinger eq.

A 'qubit' is a 2-level system.

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$$
$$|1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

A general state is $\alpha|0\rangle + \beta|1\rangle$, i.e. in general. ($\alpha,\beta\in\mathbb{C}$)

$$|\psi\rangle = \left(\cos\left(\tfrac{\theta}{2}\right), e^{i\phi}\sin\left(\tfrac{\theta}{2}\right)\right), \quad 0\le\theta\le\pi,\ 0\le\phi<2\pi$$

Proof idea

$$|\langle\psi|\psi\rangle|^2 = 1 \Rightarrow |\alpha|^2+|\beta|^2=1 \Rightarrow |\alpha|\le 1$$

$|\alpha| = \cos(\theta)$, $|\beta| = \sin(\theta)$

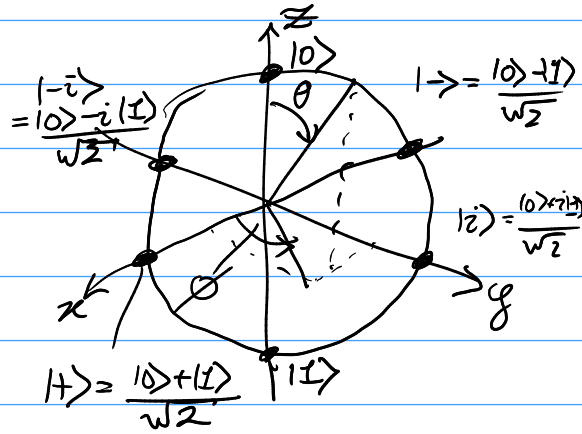
$$|\psi\rangle = \left(e^{i\gamma}\cos(\theta), e^{i\delta}\sin(\theta)\right)$$
$$= e^{i\gamma}\cdot\left(\cos(\theta), e^{i\phi}\sin(\theta)\right) \quad \phi=\delta-\gamma$$

θ, ϕ domain, think about 1-1 mapping.

A general qubit state

$$|\psi\rangle = \left(\cos\left(\frac{\theta}{2}\right),\ e^{i\phi}\sin\left(\frac{\theta}{2}\right)\right)$$

$$0 \le \theta \le \pi,\quad 0 \le \phi \le 2\pi$$



Measurement probs. $|\cdot|^2$

$$P_0 = |\alpha|^2 = \cos^2\left(\frac{\theta}{2}\right)$$
$$P_1 = |\beta|^2 = \sin^2\left(\frac{\theta}{2}\right)$$

$\cos^2(\pi/4) \sim 85\%$

$\Rightarrow$ All states map to (θ, ϕ) on Bloch sphere

$\Rightarrow$ T.E. is a rotation

$d|\psi\rangle/dx = \psi$

T.E. is given by the S.E.

$$i\hbar \frac{\partial}{\partial t}|\psi\rangle = H|\psi\rangle \Rightarrow |\psi(t)\rangle = e^{-\frac{i}{\hbar}\int_0^t H(t)\,dt}|\psi_0\rangle$$

If time indep.

$$|\psi(t)\rangle = e^{-i/\hbar\, H\cdot t}|\psi_0\rangle$$

What can H look like?

$\begin{pmatrix} \cdot & \cdot \\ \cdot & \cdot \end{pmatrix}$ 2×2 $\therefore$ 2D Hilbert space.

$\begin{pmatrix} a & x-iy \\ x+iy & b \end{pmatrix}$ $a, b \in \mathbb{R}$, $x, y \in \mathbb{R}$

$\begin{pmatrix} I+z & x-iy \\ x+iy & I-z \end{pmatrix}$ $I = \frac{a+b}{2}$, $z = \frac{a-b}{2}$

So ANY qubit (2-Level system) Hamiltonian looks like

$$H = I + X\cdot\sigma_x + Y\cdot\sigma_y + Z\cdot\sigma_z \equiv I - \sigma\cdot\hat{r}$$

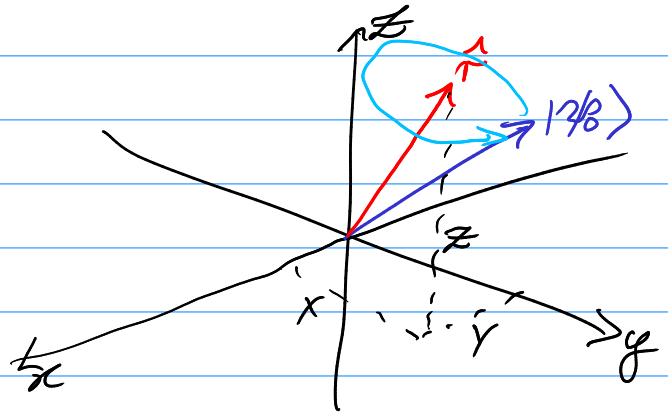
$$I=\begin{pmatrix}1&0\\0&1\end{pmatrix} \quad \sigma_x=\begin{pmatrix}0&1\\1&0\end{pmatrix} \quad \sigma_y=\begin{pmatrix}0&-i\\i&0\end{pmatrix} \quad \sigma_z=\begin{pmatrix}1&0\\0&-1\end{pmatrix}$$

What is the significance of $\hat{r}=(X,Y,Z)$?

$$R_{\hat{n}}(\varphi) = e^{-\frac{i\varphi}{2}\cdot(\sigma\cdot\hat{n})}$$ — Proof in assign.

Most important result

$$H = I + X\sigma_x + Y\sigma_y + Z\sigma_z$$



$$|\psi_0\rangle = \cos(\tfrac{\theta}{2})|0\rangle + e^{i\varphi}\sin(\tfrac{\theta}{2})|1\rangle$$

$$\hat{r}=(X,Y,Z)$$

$$\omega = \frac{2}{\hbar}\sqrt{X^2+Y^2+Z^2} = \frac{2}{\hbar}\cdot|\hat{r}|$$

$\Rightarrow$ A T.L.S./qubit's job is to keep rotating the qubit state about $\hat{r}=(X,Y,Z)$ at the Larmour precession frequency ω

$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\quad e^{A} = \sum \frac{A^n}{n!}$

$A = PDP^{-1}$

$A^n = PDP^{-1}PDP^{-1}\cdots = PD^nP^{-1}$

Some rotations...

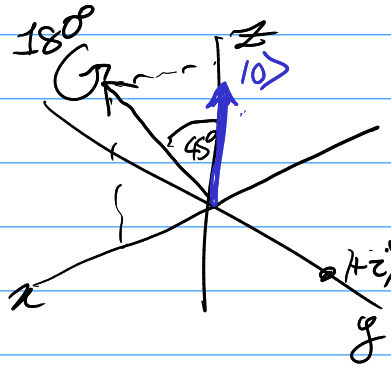
$$R_z(\varphi) = e^{-i\frac{\varphi}{2}\sigma_z} = \begin{pmatrix} e^{-i\varphi/2} & \\ & e^{+i\varphi/2} \end{pmatrix} = e^{-i\frac{\varphi}{2}}\begin{pmatrix} 1 & \\ & e^{i\varphi} \end{pmatrix}$$

$$R_x(\varphi) = e^{-i\frac{\varphi}{2}\sigma_x} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} e^{-i\varphi/2} & \\ & e^{i\varphi/2} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos(\varphi/2) & -i\sin(\varphi/2) \\ -i\sin(\varphi/2) & \cos(\varphi/2) \end{pmatrix}$$

$X_\pi = R_x(\pi) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot -i$

$X_\pi |0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$



$H \sim \sigma_x + \sigma_z$

$\frac{1}{\sqrt{2}}\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}\right)$

$= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$h|0\rangle = |+\rangle$
$h|+\rangle = |0\rangle$
$h|+i\rangle = |-i\rangle$

Note e-vals of $\sigma_x, \sigma_y, \sigma_z$ are ± 1. $\Rightarrow$ coeffs. of $\frac{1}{2}$ energy splitting.

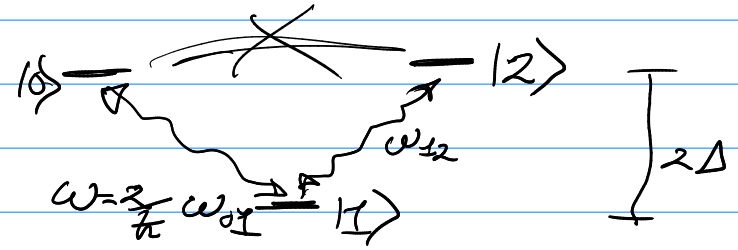
$H = B_z\sigma_z$

e-vals. $\pm B_z$

For N-level systems

$\Rightarrow$ on-diags give ϵ-splitting

$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array}\begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{pmatrix} \Delta & \omega_{01} & 0 \\ \omega_{01} & -\Delta & \omega_{12} \\ 0 & \omega_{12} & \Delta \end{pmatrix} \end{array}$$



2-Qubit Gates

N-qubit states

- Each qubit has 2D
- N qubits have 2^N dimensions
- No unique basis (eg./ 2e⁻ spin qubits use singlet-triplet basis)
- Typically use measurement/computational basis.

$|00\rangle, |01\rangle, |10\rangle, |11\rangle$

(

Masking tensor prods. ie./ $|00\rangle = |0\rangle \otimes |0\rangle$

How would an X_π gate look like if only applied to 2nd qubit.

$$\begin{array}{c} \\ 00 \\ 01 \\ 10 \\ 11 \end{array} \begin{array}{c} \begin{array}{cccc} 00 & 01 & 10 & 11 \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{array} \cdot \begin{pmatrix} \bullet \end{pmatrix} \begin{array}{c} 00 \\ 01 \\ 10 \\ 11 \end{array} \qquad \begin{array}{l} 00 \to 01 \\ 01 \to 00 \\ 10 \to 11 \\ 11 \to 10 \end{array}$$

Tedious, but a pattern emerges. It's the Kronecker/tensor product of matrices

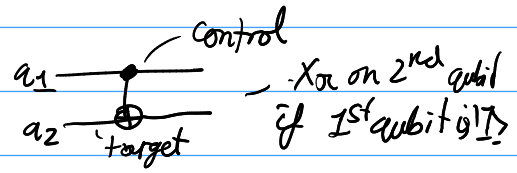
$$A \otimes B = \begin{pmatrix} A_{11}B & A_{12}B \\ A_{21}B & A_{22}B \end{pmatrix} \ldots$$

$$\mathbb{I}_2 \otimes X_\pi \equiv \mathbb{I}_2 \otimes \sigma_x = \begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}$$

So multiqubit gate matrices can be constructed from tensor products. But not all!

colloquially called cNOT

Controlled X_π - ie / CX



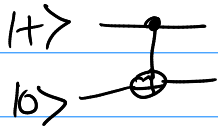
$$CX = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$

(I = identity, σ_x)

$|00\rangle \rightarrow |00\rangle$
$|01\rangle \rightarrow |01\rangle$
$|10\rangle \rightarrow |11\rangle$
$|11\rangle \rightarrow |10\rangle$

control qubit

let's feed in $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \equiv |+\rangle$ into control. $|0\rangle$ into target.



$$\rightarrow \frac{1}{\sqrt{2}} \cdot (|00\rangle + |11\rangle)$$

('Bell state')

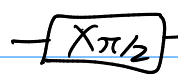
But can Bell State be written as a $\otimes$ of 2 separate qubit states? NO. It is an 'entangled state'

ex// find inputs to get other 3 'Bell basis' states $|10\rangle \pm |01\rangle$, $|00\rangle - |11\rangle$

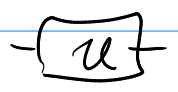
There is a zoo of gates.



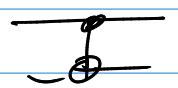
π rotation. w.r.t. x-axis



$\pi/2$ - " " " "



any " " " any "


CX or CNOT



CZ like CX but about z axis



CU (controlled unitary).

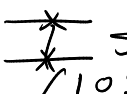
CROT(φ) controlled x axis rot. angle φ

CPHASE(φ) - " " " z axis.

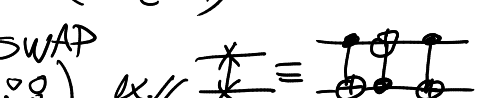

CCX or Toffoli X rot on 3rd qubit if first 2 qubits are both 1

$Z_{\pi/2} \equiv \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$ - S-gate

$Z_{\pi/4} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$ - T-gate.


SWAP

$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

ex// 

There is no copy gate. No cloning thm.

$$|\psi\rangle \otimes |0\rangle \not\rightarrow |\psi\rangle \otimes |\psi\rangle$$

